

PowerRanger

A Server-Level Power Controller for Grid-Responsive Datacenters

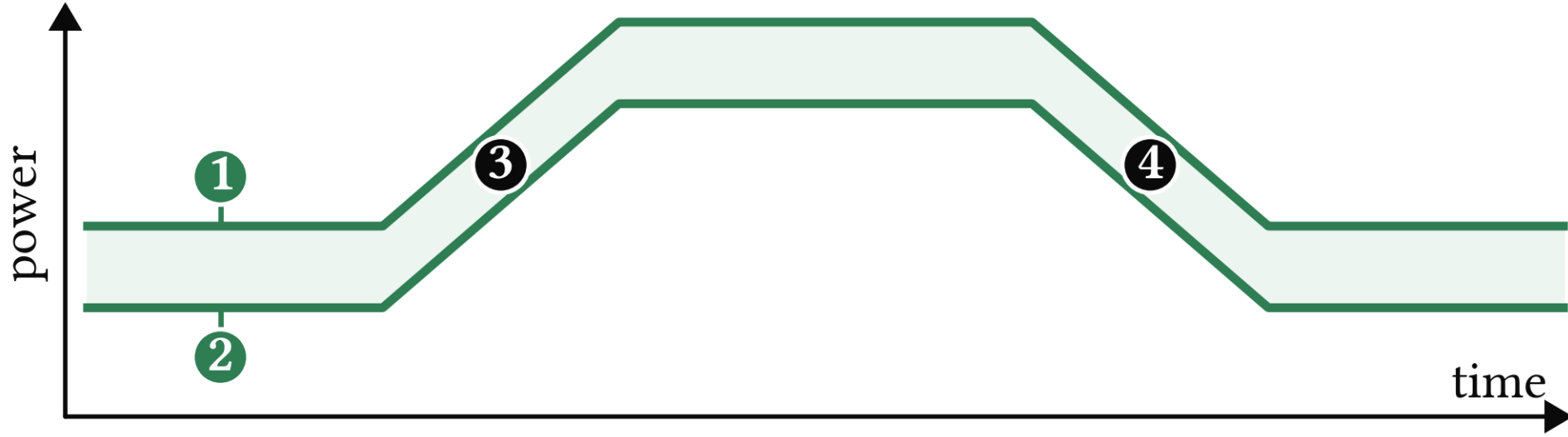
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Motivation

Synchronized AI jobs swing a datacenter's power between peak and near-idle in seconds. These swings strain turbines, and grid operators now limit how fast and how far that load may change.

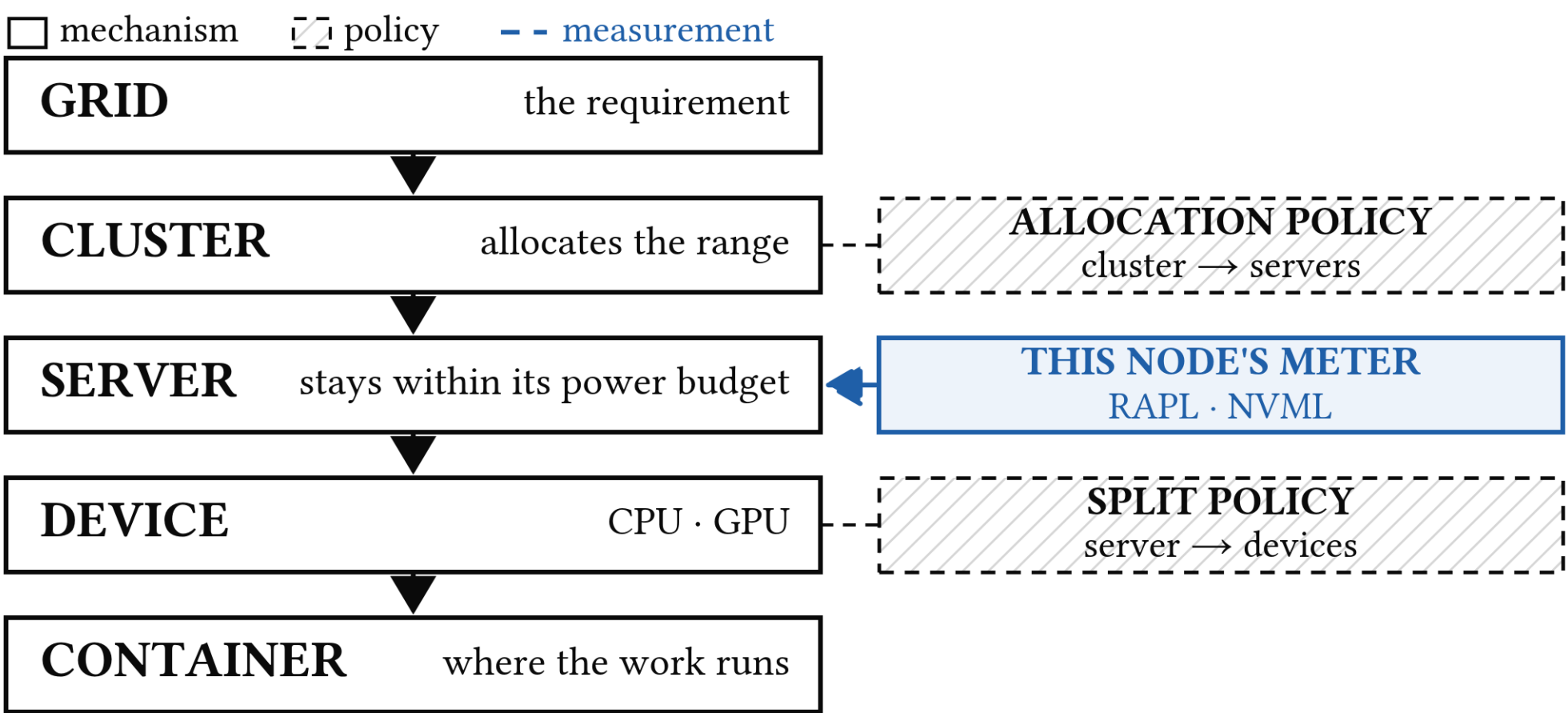
Can we respect the grid's constraints while keeping the cluster's throughput?

The Power Range Abstraction

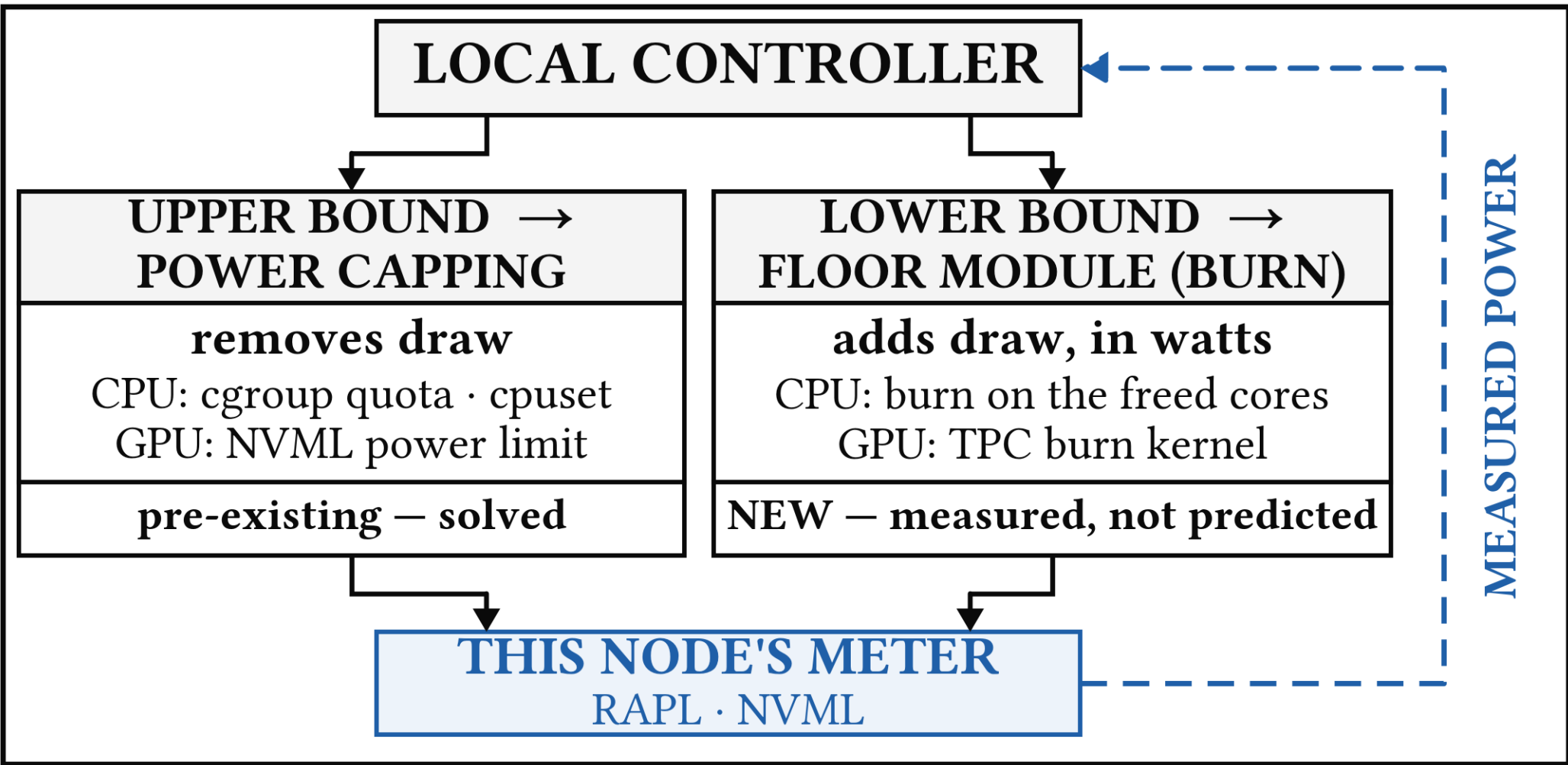


- ① set_upper_bound
- ② set_lower_bound
- ③ set_ramp_up_rate
- ④ set_ramp_down_rate

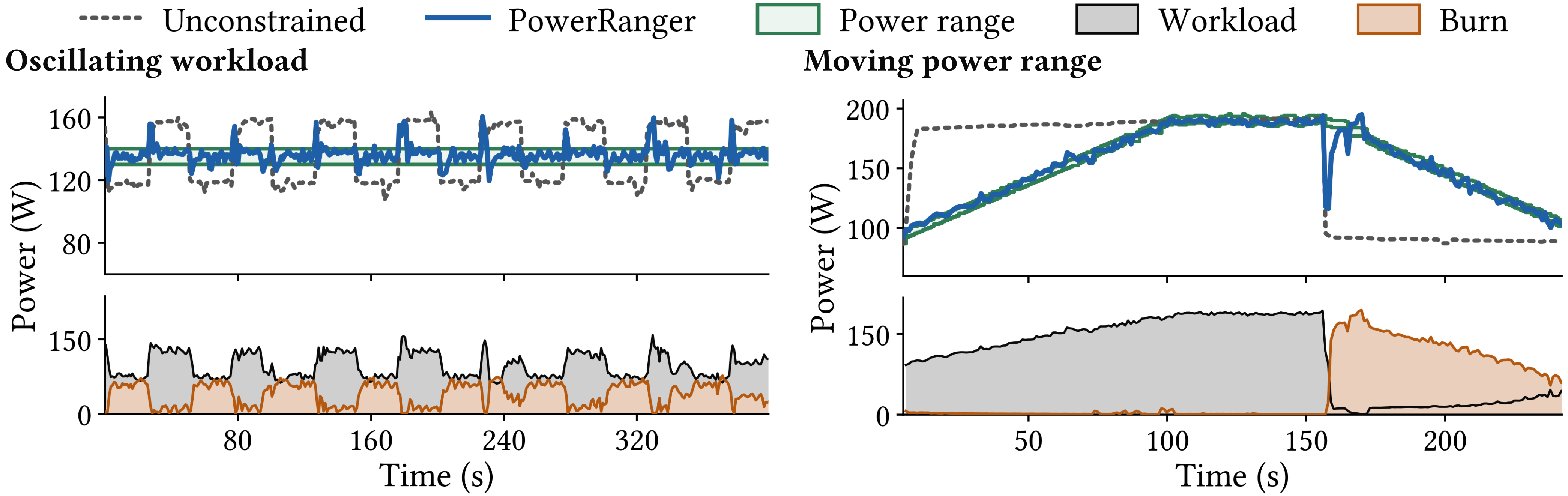
System Overview



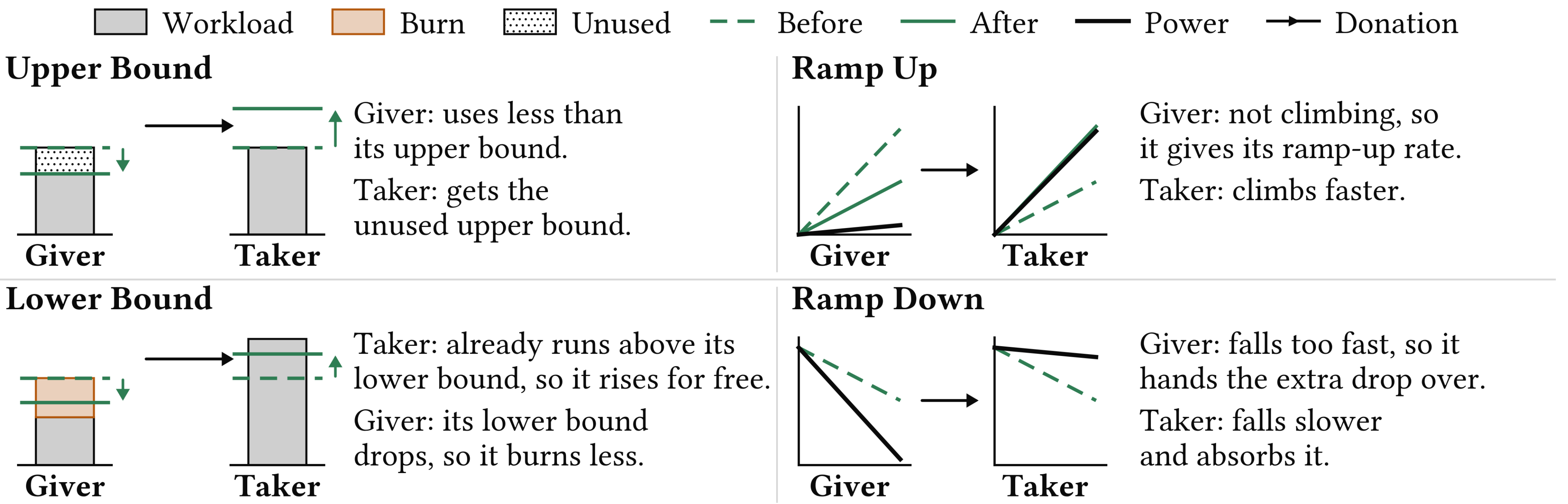
Server-Level Mechanisms



Preliminary Results



Power Allocation Policy



Takeaways and Future Work

A server-level power controller
PowerRanger holds a moving power range on each node, closing the loop on the node's own meter.

The floor module (burn) is new
Capping only removes power draw. Burn adds measured watts, so the floor holds when work collapses.

Next: large scale, real apps
Multi-server tests at cluster scale, then real AI and non-AI applications on CPU and GPU.